

Task 3: Deep-Dive Sales Analysis & Interactive Dashboarding

Project Title	Deep-Dive Sales Analysis & Interactive Dashboarding
Submitted By	Mohan Teja Palli
Tool Used	Power BI Desktop
Audience	Internship Supervisor / Business Stakeholders
Date	June 2025

1. Objective

This project aims to perform an in-depth sales analysis using Power BI Desktop and deliver interactive dashboards that provide actionable business insights. The key objectives are:

- **Evaluate regional sales distribution** across East, West, North, and South regions.
 - **Explore product rankings** to identify top-performing and underperforming items.
 - **Build interactive Power BI dashboards** to monitor business performance across categories, regions, and time periods.
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2. Dataset Overview

The dataset used for this analysis contains retail transactional data with the following fields:

Field	Description
Order ID	Unique identifier for each order
Order Date	Date the order was placed
Product	Name of the product sold
Category	Product category (Fashion, Furniture, Home Appliances, Stationery, Technology)
Region	Geographic region (East, West, North, South)
Sales	Revenue generated from the order
Profit	Profit earned from the order

Tool Used: Power BI Desktop – used for data modeling, DAX measures, and interactive visual creation.

3. Key Performance Indicators (KPIs)

KPI	Value
Total Orders	40
Total Sales	510K
Total Profit	104K
Average Sales per Order	12.75K

4. Dashboard Page 1 – Sales & Profit Performance

-Dashboard Components

- **KPI Cards** – Display Total Sales (510K), Total Profit (104K), Total Orders (40), and Average Sales per Order (12.75K) at a glance.
- **Category-Wise Sales (Pie Chart)** – Visualizes the share of sales across all five product categories.
- **Region-Wise Profit (Bar Chart)** – Compares profit contribution across East, North, South, and West regions.
- **Monthly Sales Trend (Line Chart)** – Tracks sales performance from January through June 2025.
- **Top Products by Sales (Bar Chart)** – Ranks individual products by total revenue generated.

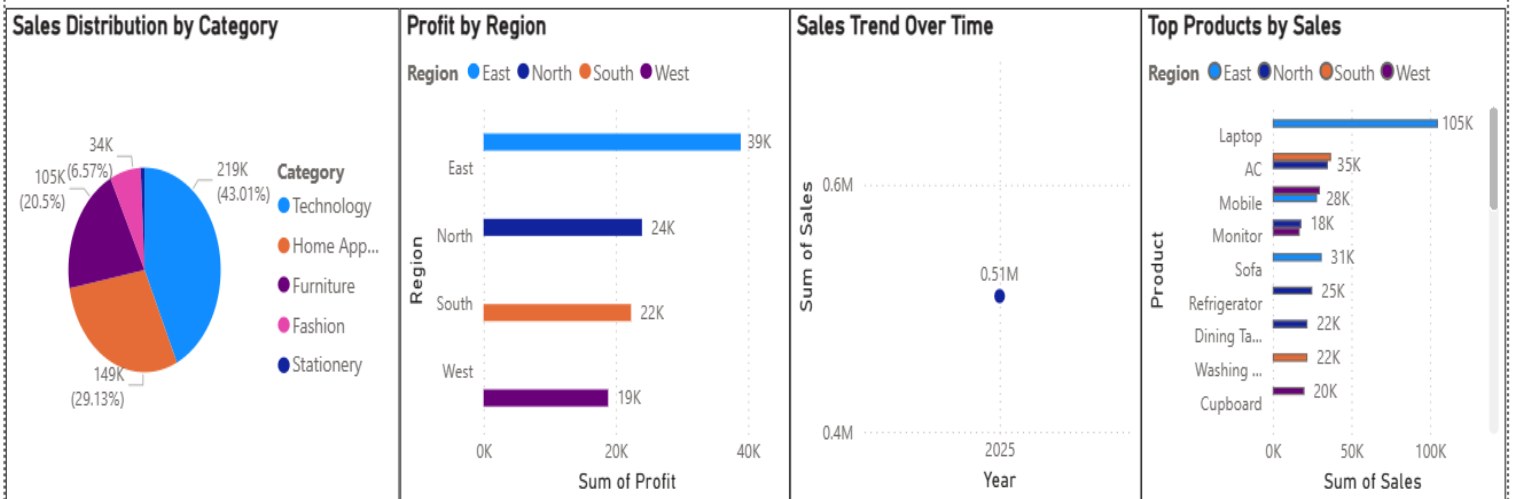
Key Findings

- **Technology dominates sales**, accounting for the largest share (43%) of total revenue across all categories.
 - **East region leads profit**, generating the highest profit (39K) among all four regions.
 - **Seasonal variation observed** – sales peaked in February and experienced a notable dip in June, suggesting demand fluctuations tied to seasonal or promotional factors.
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Sales & Profit Performance Dashboard

Total Sales	Total Profit	Total Orders	Average Sales per Order
510K	104K	40	12.75K
Sum of Sales	Sum of Profit	Count of OrderID	Average of Sales

Category ☐ Fashion ☐ Furniture ☐ Home Appliances ☐ Stationery ☐ Technology	Region ☐ East ☐ North ☐ South ☐ West	OrderDate 05-01-2025 12-06-2025
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5. Dashboard Page 2 – Sales & Profit Deep Dive

Dashboard Components

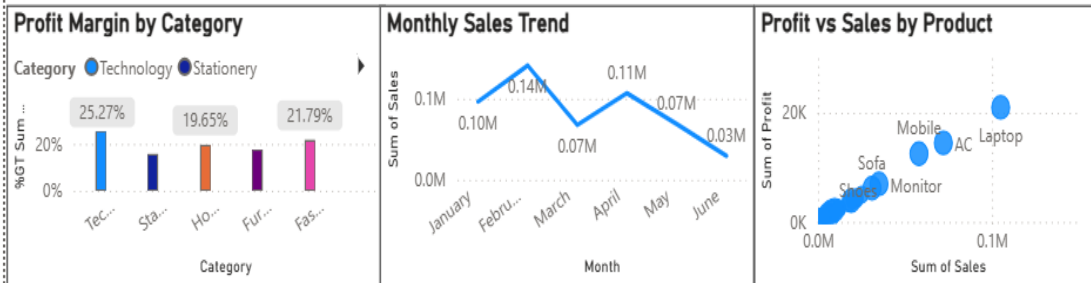
- **Profit Margin by Category (Bar Chart)** — Compares percentage profit margins: Technology (25.27%), Stationery (19.65%), Home Appliances, Furniture, and Fashion (21.79%).
- **Monthly Sales Trend (Line Chart)** – Detailed month-over-month sales movement from January to June 2025.
- **Profit vs Sales Scatter Plot** — Maps each product on profit-sales axes to identify high-value vs. low-value items.
- **Sales by Region & Category (Matrix Table)** – Cross-tabulates sales data across all regions and product categories.
- **Category Performance by Month (Clustered Bar Chart)** – Shows monthly sales trends broken down by category.

Key Findings

- **Technology products consistently dominate** across all months and regions, confirming category leadership.
 - **Strong sales-profit correlation** observed in the scatter plot – higher-sales products (Laptop, Mobile) also yield higher profits.
 - **East region remains consistent** as the top-performing region regardless of category or time period.
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Sales & Profit Deep Dive Dashboard

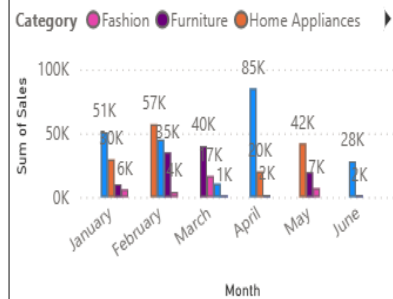
Category ☐ Fashion ☐ Furniture ☐ Home Appliances ☐ Stationery ☐ Technology	Region ☐ East ☐ North ☐ South ☐ West	OrderDate 05-01-2025 12-06-2025
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Sales by Region and Category

Region	Fashion	Furniture	Home Appliances	Stationery	Technology	Total
West	4500	30500	8000	1500	47800	92300
South	7300	18000	63500	1000	19000	108800
North	9700	22000	65000	700	19500	116900
East	12000	34000	12000	850	133000	191850
Total	33500	104500	148500	4050	219300	509850

Sum of Sales by Month and Category



6. Interactive Features

The dashboards include the following interactive capabilities:

- **Dynamic Slicers** – Users can filter visuals by Category, Region, and Order Date range simultaneously.
 - **Cross-Filtering** – Selecting a segment in one visual automatically filters all other visuals on the page for contextual analysis.
 - **Drill-Down Analysis** – Enables users to explore data from category level down to individual product level.
 - **Real-Time KPI Updates** – All KPI cards update dynamically based on active filter selections.
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7. Business Insights

- Technology is the strongest contributor to both revenue and profit, making it the highest-priority category for inventory and marketing investment.
- East region consistently outperforms other regions in both sales and profit, suggesting a stronger customer base or more effective distribution channels.
- **High-sales products drive profit** – items such as Laptops, Mobiles, and ACs show a strong positive correlation between revenue and profitability.
- Interactive dashboards support quick decision-making by enabling stakeholders to filter, compare, and drill down into data without requiring technical expertise.

8. Conclusion

The Power BI dashboard successfully delivers actionable insights from the sales dataset by:

- Providing a **comprehensive view of KPIs** through dynamic cards and interactive filters.
- Enabling **regional and category-level comparisons** to identify performance gaps and opportunities.
- Supporting **trend analysis** across months to detect seasonal patterns and anomalies.
- Empowering **data-driven decision-making** for business stakeholders through an intuitive, self-service analytics interface.

This project demonstrates the effective use of Power BI Desktop for end-to-end business intelligence — from data ingestion and modeling to visual storytelling and interactive reporting.
